Mathematical and Computational Methods

Lecture 8: Differentiation II: applications

Lecture 7 built the derivative and the rules for computing it. This lecture puts that machinery to work. We begin with the geometry it controls—the *tangent* and *normal* lines—and with four theorems that underpin everything which follows: the intermediate-value, extreme-value and mean-value theorems, and the rule for differentiating an inverse, together with *l'Hôpital's rule* for awkward limits. We then use the first and second derivatives to locate and classify the turning points of a curve and to sketch it accurately, and finally to solve *optimisation* problems—closing with *gradient descent*, the numerical idea that turns the derivative into an engine for minimisation across the sciences and data science.

Where this fits. Everything rests on Lecture 7: the derivative and its three readings, the rules, higher derivatives, and the continuity it requires. The mean-value theorem is what licenses the increasing/decreasing test of Section 2, and the extreme-value theorem is what licenses the endpoint check in the optimisation of Section 3. The curve-sketching skills feed directly into the area and volume problems of Lectures 9–10, and the optimisation and gradient-descent material is foundational for the data-science and physics students alike.

Contents

Learning objectives

By the end of this lecture you will be able to:

- Find tangent and normal lines.
- State the key theorems (IVT, EVT, MVT, inverse-function) and apply l'Hôpital's rule.
- Solve equations numerically by Newton's method.
- Locate and classify stationary points with the first- and second-derivative tests.
- Sketch curves of polynomial and rational functions using calculus.
- Solve optimisation problems and explain gradient descent.

1 Tangents, normals and the key theorems

1.1 Tangent and normal lines

The geometric reading of Lecture 7 says $f'(a)$ is the slope of the curve at $\big(a, f(a)\big)$. Two lines through that point are determined by it.

Definition 1.1 (Tangent and normal lines). At the point $\big(a, f(a)\big)$ on a differentiable curve $y = f(x)$, the *tangent line* is

$$y = f(a) + f'(a)\,(x - a),$$

and the *normal line* is the perpendicular through the same point. When $f'(a) \neq 0$ the normal has slope $-1/f'(a)$, so

$$y = f(a) - \frac{1}{f'(a)}\,(x - a).$$

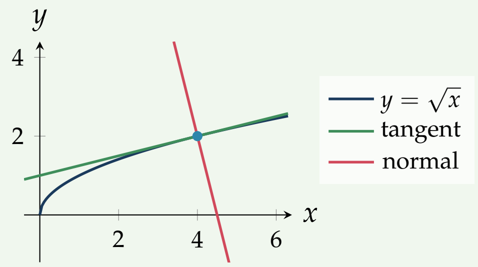
If $f'(a) = 0$ the tangent is horizontal and the normal is the vertical line $x = a$.

The normal slope is $-1/f'(a)$ because perpendicular lines have slopes whose product is -1. The normal direction is the one in which a surface pushes back on a body resting against it—the *normal force* of mechanics—which is where the name comes from. (A particle released from circular motion, by contrast, flies off along the *tangent*, not the normal.)

Example 1.1 (*Tangent and normal to $y = \sqrt{x}$*). In Lecture 7 we found the tangent to $y = \sqrt{x}$ at $x = 4$: there $f(4) = 2$ and $f'(4) = \frac{1}{4}$, giving the tangent $y = \frac{1}{4}x + 1$. The normal at the same point has slope $-1/f'(4) = -4$, so

$$y - 2 = -4\,(x - 4) \qquad \Longrightarrow \qquad y = -4x + 18.$$

Tangent and normal meet at right angles at $(4, 2)$, as their slopes $\frac{1}{4}$ and -4 confirm. With equal scales on the two axes, that right angle is visible directly.



1.2 Two theorems for continuous functions

The next two theorems cost nothing but continuity, yet they guarantee the existence of things we will need: solutions of equations, and maxima and minima.

Theorem 1.1 (Intermediate value theorem). If f is continuous on $[a, b]$ and N is any value between $f(a)$ and $f(b)$, then $f(c) = N$ for some c in (a, b).

In words, a continuous function takes every value between any two it attains—its graph cannot jump over a level. The most useful case is $N = 0$: if f is continuous and changes sign on $[a, b]$, it has a root in between.

Example 1.2 *(Locating a root).* Let $f(x) = x^3 - x - 1$. It is continuous, with $f(1) = -1 < 0$ and $f(2) = 5 > 0$. By the intermediate value theorem there is a root in $(1, 2)$. This is also the reasoning behind the remark in Lecture 1 that every odd-degree polynomial has at least one real root—its values run from $-\infty$ to $+\infty$, or the reverse if the leading coefficient is negative, so either way it must cross zero.

Theorem 1.2 (Extreme value theorem). If f is continuous on a *closed, bounded* interval $[a, b]$, then f attains an absolute maximum and an absolute minimum on $[a, b]$: there are points $c, d \in [a, b]$ with $f(d) \le f(x) \le f(c)$ for all $x \in [a, b]$.

Note. Both hypotheses are essential. On the *open* interval $(0, 1)$ the function $f(x) = x$ attains neither a maximum nor a minimum, and on $(0, 1]$ the function $1/x$ has no maximum (it runs off to ∞ near 0). The theorem is what guarantees that the optimisation problems of Section 3 actually have a best answer, and that we may find it by checking finitely many candidates.

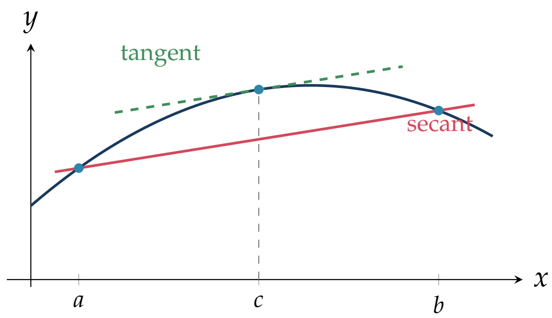
1.3 The Mean Value Theorem

Where the previous two theorems used only continuity, the mean value theorem uses differentiability, and it is the workhorse behind the curve-analysis of Section 2.

Theorem 1.3 (Mean value theorem). If f is continuous on $[a, b]$ and differentiable on (a, b), then there is a point $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Geometrically: somewhere strictly between a and b the tangent is parallel to the secant joining the endpoints—the instantaneous rate of change equals the average rate of change at least once.



The single most useful consequence concerns the *sign* of the derivative. If f' is positive throughout an interval, then for any two points the right-hand side above is positive, so f increases; and likewise for the other signs.

Corollary 1.1 (Monotonicity from the derivative). On an interval where f is differentiable:

$$f' > 0 \;\Rightarrow\; f \text{ strictly increasing}, \qquad f' < 0 \;\Rightarrow\; f \text{ strictly decreasing}, \qquad f' \equiv 0 \;\Rightarrow\; f \text{ constant}.$$

This is the precise statement behind the intuition "positive slope means going uphill," and it is the tool we use in Section 2 to read a curve's shape off its derivative.

1.4 The derivative of an inverse function

Lecture 1 built inverse functions by reflecting a graph in the line $y = x$. That reflection swaps the roles of "rise" and "run," so it should turn a slope m into $1/m$—which is exactly what happens to the derivative.

Theorem 1.4 (Derivative of an inverse). Let f be differentiable and one-to-one, and write $a = f^{-1}(b)$. If $f'(a) \neq 0$, then f^{-1} is differentiable at b with

$$\big(f^{-1}\big)'(b) = \frac{1}{f'\big(f^{-1}(b)\big)}.$$

The rise/run picture is the intuition; the chain rule of Lecture 7 makes it exact. Since f^{-1} undoes f, the two compose to the identity, $f\big(f^{-1}(x)\big) = x$. Differentiating both sides and applying the chain rule on the left gives

$$f'\big(f^{-1}(x)\big) \cdot \big(f^{-1}\big)'(x) = 1 \qquad \Longrightarrow \qquad \big(f^{-1}\big)'(x) = \frac{1}{f'\big(f^{-1}(x)\big)},$$

which is the formula above. This also explains the hypothesis $f' \neq 0$: it is exactly what stops the right-hand side dividing by zero (geometrically, a horizontal tangent of f reflects to a vertical tangent of f^{-1}, where the slope is undefined).

Example 1.3 (*Recovering the derivative of* $\ln$). The natural logarithm is the inverse of $f(x) = \mathrm{e}^x$, for which $f' = \mathrm{e}^x$. With $f^{-1}(x) = \ln x$,

$$(\ln)'(x) = \frac{1}{f'(\ln x)} = \frac{1}{\mathrm{e}^{\ln x}} = \frac{1}{x},$$

the standard derivative of Lecture 7. The same method (or, equivalently, implicit differentiation) gives the inverse-trigonometric derivatives

$$\frac{\mathrm{d}}{\mathrm{d}x} \arcsin x = \frac{1}{\sqrt{1 - x^2}}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} \arctan x = \frac{1}{1 + x^2},$$

completing the catalogue of standard derivatives begun in Lecture 7.

1.5 l'Hôpital's rule

Limits of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ are *indeterminate*: the answer depends on *how* numerator and denominator approach their limits. Differentiation resolves them.

Theorem 1.5 (l'Hôpital's rule). Suppose f and g are differentiable near a with $g' \neq 0$ there, and that $\dfrac{f(x)}{g(x)}$ has the indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ as $x \to a$. Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)},$$

provided the right-hand limit exists (or is $\pm\infty$). The point a may be finite or $\pm\infty$.

Example 1.4 ***(Two limits, one repeated).*** For the $\frac{0}{0}$ form

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} \stackrel{\text{H}}{=} \lim_{x\to 0} \frac{\sin x}{2x} \stackrel{\text{H}}{=} \lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2},$$

applying the rule twice (the first application is still $\frac{0}{0}$, so we check and differentiate again). For the $\frac{\infty}{\infty}$ form,

$$\lim_{x\to\infty} \frac{\ln x}{x} \stackrel{\text{H}}{=} \lim_{x\to\infty} \frac{1/x}{1} = 0,$$

which says $\ln x$ grows more slowly than x.

Note. l'Hôpital applies *only* to indeterminate forms. Differentiating top and bottom of a limit that is not $\frac{0}{0}$ or $\frac{\infty}{\infty}$ gives the wrong answer: always confirm the form first. Other indeterminate forms ($0 \cdot \infty$, $\infty - \infty$, 1^∞, $\dots$) can usually be rearranged into one of these two.

1.6 Newton's method

The intermediate value theorem *locates* a root—it tells us a continuous, sign-changing f has a zero in $(1, 2)$—but it does not say *where*. To pin the root down we put the tangent line to work. A tangent is the curve's best straight-line proxy near the point of contact, so if our current estimate is close to a root, the tangent there crosses the axis very close to where the curve does. Following the tangent to the axis turns a rough estimate into a sharper one; repeating the step drives it home.

Concretely, the tangent to $y = f(x)$ at the current estimate x_n is

$$y = f(x_n) + f'(x_n)\,(x - x_n).$$

Setting $y = 0$ and solving for x gives the point where this tangent meets the axis, which we take as the next estimate x_{n+1}:

$$0 = f(x_n) + f'(x_n)\,(x_{n+1} - x_n) \qquad \Longrightarrow \qquad \boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}, \qquad f'(x_n) \neq 0.$$

This is *Newton's method* (or the Newton–Raphson method): start from a guess x_0 and iterate.

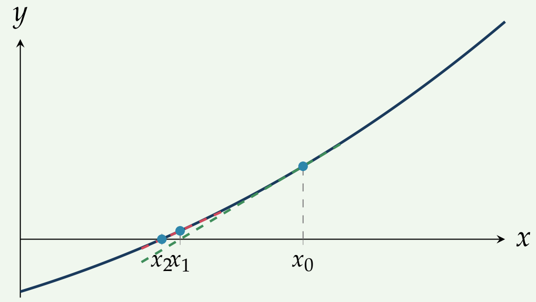
Example 1.5 ***(Newton's method on $x^3 - x - 1$).*** We return to the root of $f(x) = x^3 - x - 1$ that the intermediate value theorem placed in $(1, 2)$. Here $f'(x) = 3x^2 - 1$, so the iteration is

$$x_{n+1} = x_n - \frac{x_n^3 - x_n - 1}{3x_n^2 - 1}.$$

Starting from $x_0 = 1.5$:

n	x_n	$f(x_n)$
0	1.500000	0.875000
1	1.347826	0.100682
2	1.325200	0.002058
3	1.324718	0.000001
4	1.324718	0.000000

The estimates settle on $x \approx 1.324718$, the real root to six decimal places, in three or four steps. Notice the number of correct digits roughly *doubles* each step ($0.15 \to 0.02 \to 0.0005 \to \dots$ in the error): this rapid, "quadratic" convergence is what makes the method so useful. The geometry is shown below—each step slides down the tangent to the axis.



Note (When Newton works—and when it does not). Newton's method is fast but not foolproof. Started near a simple root (one where $f' \neq 0$) it converges rapidly, but a poor starting guess can send it off to a different root or away altogether, and if $f'(x_n)$ is near zero the step $f(x_n)/f'(x_n)$ becomes huge and the iteration can jump wildly. The intermediate value theorem offers the slow-but-safe complement: it is guaranteed to trap a root once a sign change is found, where Newton trades that guarantee for speed. The same "step using the derivative" idea returns in Section 3 as gradient descent—there to minimise rather than to solve $f = 0$.

2 Critical points and curve sketching

2.1 Critical points and the first-derivative test

A high or low point of a graph has a horizontal tangent, so the search for the peaks and troughs of a function starts by setting its derivative to zero.

Definition 2.1 (Critical and stationary points). A *critical point* of f is a value $x = c$ in its domain where $f'(c) = 0$ or $f'(c)$ fails to exist. A point where $f'(c) = 0$ is also called a *stationary point*. The corresponding point $(c, f(c))$ is a candidate for a local maximum or minimum.

Theorem 2.1 (Fermat's theorem). If f has a local maximum or minimum at an interior point c of its domain and $f'(c)$ exists, then $f'(c) = 0$. Consequently an extremum can occur only at a critical point or at an endpoint—which is why checking those finitely many points suffices.

By the monotonicity corollary, f increases where $f' > 0$ and decreases where $f' < 0$, so the behaviour of a stationary point is decided by how the sign of f' changes across it.

Theorem 2.2 (First-derivative test). Let c be a stationary point of f. As x increases through c:

$$f' : + \to - \;\Rightarrow\; \text{local maximum}, \qquad f' : - \to + \;\Rightarrow\; \text{local minimum},$$

and if f' does *not* change sign, c is neither (the curve levels off and continues the same way—a horizontal inflection).

2.2 Concavity and the second-derivative test

The second derivative records how the slope is changing, which is the same as the way the curve bends.

Definition 2.2 (Concavity and inflection). On an interval, f is *concave up* (holds water) where $f'' > 0$ and *concave down* where $f'' < 0$. A point where the concavity changes—so f'' changes sign—is an *inflection point*.

At a stationary point the concavity decides the type directly, which is usually quicker than tracking the sign of f' on both sides.

Theorem 2.3 (Second-derivative test). Suppose $f'(c) = 0$. Then

$$f''(c) > 0 \;\Rightarrow\; \text{local minimum at } c, \qquad f''(c) < 0 \;\Rightarrow\; \text{local maximum at } c.$$

If $f''(c) = 0$ the test is inconclusive and one falls back on the first-derivative test.

The test is just the first-derivative test in disguise. Suppose $f'(c) = 0$ and $f''(c) > 0$. Then f'' is the derivative of f', so $f''(c) > 0$ says f' is *increasing* as x passes through c; since $f'(c) = 0$, the slope f' therefore runs from negative to positive across c. By the first-derivative test that is a local minimum. The case $f''(c) < 0$ is the mirror image, with f' decreasing through zero, hence a maximum. When $f''(c) = 0$ this argument gives no information—f' need not change sign—which is precisely why the test is then inconclusive: compare $f(x) = x^4$ (a minimum at 0) with $f(x) = x^3$ (no extremum), both of which have $f'(0) = f''(0) = 0$.

Example 2.1 *(Classifying stationary points).* For $f(x) = x^3 - 3x$, $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$, so the stationary points are $x = \pm 1$. With $f''(x) = 6x$,

$$f''(1) = 6 > 0 \;\Rightarrow\; \text{local minimum at } (1, -2),$$

$$f''(-1) = -6 < 0 \;\Rightarrow\; \text{local maximum at } (-1, 2).$$

The second-derivative test settles both in one line each.

Remark (The sigmoid, revisited). Lecture 7 found $\sigma'(x) = \sigma(x)\big(1 - \sigma(x)\big)$ for the logistic function. Since $0 < \sigma < 1$, this is positive for every x, so by the monotonicity corollary σ is strictly increasing everywhere—it never turns back, which is why it makes a good probability. Differentiating again gives $\sigma'' = \sigma'(1 - 2\sigma)$, which vanishes only at $\sigma = \frac{1}{2}$, i.e. $x = 0$: the sigmoid has a single inflection there, switching from concave up to concave down.

2.3 Curve sketching

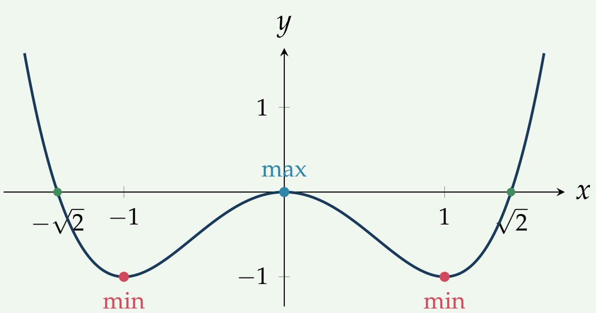
Together, f, f' and f'' give enough information for an accurate sketch without plotting many points. A systematic pass collects:

1. **Domain and symmetry** — where f is defined; even ($f(-x) = f(x)$) or odd ($f(-x) = -f(x)$) symmetry from Lecture 1.
2. **Intercepts** — $f(0)$ for the y-axis; solve $f(x) = 0$ for the x-axis.
3. **Asymptotes** — for rational functions, vertical asymptotes at the zeros of the denominator and horizontal asymptotes from the limits at infinity of Lecture 7.
4. **Increase/decrease and extrema** — the sign of f' and the stationary points it locates.
5. **Concavity and inflections** — the sign of f''.

Example 2.2 ***(A full sketch:*** $f(x) = x^4 - 2x^2$***).***

- *Domain and symmetry.* Defined on all of $\mathbb{R}$; even, since $f(-x) = f(x)$, so the graph is symmetric about the y-axis.
- *Intercepts.* $f(0) = 0$, and $f(x) = x^2(x^2 - 2) = 0$ gives $x = 0$ and $x = \pm\sqrt{2}$.
- *Increase/decrease.* $f'(x) = 4x^3 - 4x = 4x(x-1)(x+1)$, with stationary points $x = -1, 0, 1$. The sign of f' runs $-, +, -, +$ across these, so f decreases on $(-\infty, -1)$, increases on $(-1, 0)$, decreases on $(0, 1)$ and increases on $(1, \infty)$.
- *Classification.* $f''(x) = 12x^2 - 4$. Then $f''(\pm 1) = 8 > 0$, giving local minima at $(\pm 1, -1)$, and $f''(0) = -4 < 0$, giving a local maximum at $(0, 0)$.
- *Concavity.* $f'' = 0$ at $x = \pm\frac{1}{\sqrt{3}}$, with $f'' > 0$ outside and $f'' < 0$ between: the curve is concave up for $|x| > \frac{1}{\sqrt{3}}$ and concave down between, with inflection points at $x = \pm\frac{1}{\sqrt{3}}$.

The resulting "W" shape is drawn below, with the two minima, the central maximum and the x-intercepts marked.



A rational function adds the one feature the polynomial above lacks—asymptotes—and this is where the limits of Lecture 7 earn their keep.

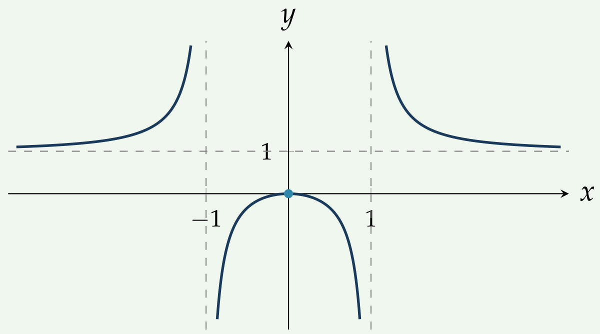
Example 2.3 (*A rational sketch:* $f(x) = \dfrac{x^2}{x^2-1}$).

- *Domain and symmetry.* The denominator $x^2 - 1 = (x-1)(x+1)$ vanishes at $x = \pm 1$, so the domain is $\mathbb{R} \setminus \{-1, 1\}$; the function is even.
- *Intercepts.* $f(0) = 0$ is the only intercept.
- *Asymptotes.* Near $x = \pm 1$ the denominator $\to 0$ while the numerator does not, so $x = \pm 1$ are *vertical* asymptotes. For the behaviour at infinity, divide through by x^2:

$$\lim_{x\to\pm\infty} \frac{x^2}{x^2-1} = \lim_{x\to\pm\infty} \frac{1}{1-1/x^2} = 1,$$

  so $y = 1$ is a *horizontal* asymptote.
- *Increase/decrease.* By the quotient rule $f'(x) = \dfrac{-2x}{(x^2-1)^2}$, which is zero only at $x = 0$. Since the denominator is positive, $f' > 0$ for $x < 0$ and $f' < 0$ for $x > 0$: the only stationary point is a local maximum at $(0, 0)$.
- *Concavity.* Differentiating again, $f''(x) = \dfrac{2(3x^2+1)}{(x^2-1)^3}$, whose sign is that of $(x^2-1)^3$: the curve is concave *up* on the outer branches ($|x| > 1$) and concave *down* in the middle ($|x| < 1$), with no inflection (the sign change occurs only across the asymptotes).

Between the vertical asymptotes the curve opens *downward* to $-\infty$, with its peak the local maximum at the origin $(0, 0)$; on the outer branches it descends from $+\infty$ towards the horizontal asymptote $y = 1$, staying above it. This gives the three-piece shape below.



3 Optimisation

3.1 Solving optimisation problems

Optimisation asks for the largest or smallest value of some quantity. The calculus is the easy part; the skill is translating a description into a function of *one* variable and identifying the interval on which it is to be optimised.

Note (A procedure for optimisation).

1. Name the quantity Q to be optimised and the variables it depends on.
2. Use the constraints to write Q as a function of a *single* variable, and state the domain (the physically meaningful interval).
3. Find the critical points by solving $Q' = 0$.
4. Identify the optimum among the critical points and, on a closed interval, the *endpoints*; classify with the second-derivative test where convenient.

The endpoint check in step 4 is not optional: by the extreme value theorem a continuous function on a closed interval attains its extremes *somewhere*, but that somewhere may be a boundary rather than a stationary point. Checking the (finitely many) critical points and endpoints is therefore guaranteed to find the answer.

Example 3.1 ***(The most economical can).*** A closed cylindrical can must hold a volume of 1000 cm^3. What dimensions use the least metal?

The quantity to minimise is the surface area A of a cylinder of radius r and height h,

$$A = \underbrace{2\pi r^2}_{\text{two ends}} + \underbrace{2\pi r h}_{\text{side}}.$$

The volume constraint $\pi r^2 h = 1000$ fixes $h = \dfrac{1000}{\pi r^2}$, which removes h and leaves a function of r alone:

$$A(r) = 2\pi r^2 + 2\pi r \cdot \frac{1000}{\pi r^2} = 2\pi r^2 + \frac{2000}{r}, \qquad r > 0.$$

Setting the derivative to zero,

$$A'(r) = 4\pi r - \frac{2000}{r^2} = 0 \qquad \Longrightarrow \qquad r^3 = \frac{500}{\pi}, \qquad r = \left(\frac{500}{\pi}\right)^{1/3} \approx 5.42\,\text{cm}.$$

Since $A''(r) = 4\pi + \dfrac{4000}{r^3} > 0$ for all $r > 0$, this critical point is a minimum—and as $r \to 0$ or $r \to \infty$ the area grows without bound, so it is the global minimum. The corresponding height is

$$h = \frac{1000}{\pi r^2} = 2r,$$

so the most economical can is exactly as tall as it is wide. (The tidy result $h = 2r$ follows on substituting $r^3 = 500/\pi$ into $h = 1000/(\pi r^2)$.)

3.2 Gradient descent

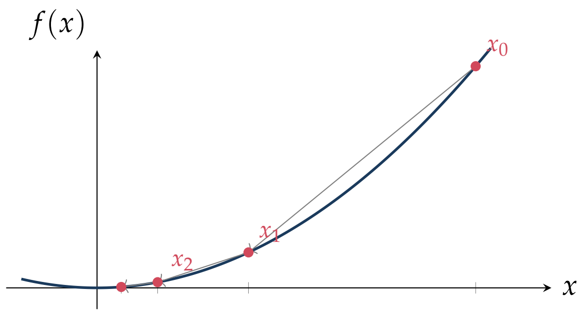
Solving $f'(x) = 0$ by hand works for the textbook functions above, but in data science a quantity to be minimised—an *error* or *loss* L—may depend on thousands of parameters through a formula far too tangled to solve exactly. We instead minimise *numerically*, and the derivative shows the way.

Recall the third reading of the derivative from Lecture 7: $f'(x)$ is a *sensitivity*, the direction and rate in which f responds to a nudge in x. If $f'(x) > 0$ then f rises to the right, so to *decrease* f we should step to the *left*—against the derivative. This is the whole idea.

Note (Gradient descent). To minimise f, start from a guess x_0 and repeatedly step against the derivative,

$$x_{n+1} = x_n - \eta\, f'(x_n),$$

where the *learning rate* $\eta > 0$ sets the step size. The iterates move downhill towards a stationary point, where $f' = 0$ and the steps stop.



Example 3.2 (*Descending a parabola*). Minimise $f(x) = x^2$, for which $f'(x) = 2x$, starting at $x_0 = 2.5$ with $\eta = 0.3$. The update is $x_{n+1} = x_n - 0.3\,(2x_n) = 0.4\,x_n$, so

$$x_0 = 2.5, \quad x_1 = 1.0, \quad x_2 = 0.4, \quad x_3 = 0.16, \ \ldots \ \to 0,$$

homing in on the minimum at $x = 0$ (shown above). The steps shrink as the slope flattens—the method naturally slows as it nears the bottom.

How small is "too small" and how large is "too large"? A quadratic model makes the threshold exact. Consider gradient descent applied to $f(x) = \frac{1}{2}k(x - x^*)^2$ with $k > 0$, a parabola with its minimum at x^*; then $f'(x) = k(x - x^*)$ and the update $x_{n+1} = x_n - \eta f'(x_n)$ becomes

$$x_{n+1} - x^* = (x_n - x^*) - \eta\, k\,(x_n - x^*) = (1 - \eta k)\,(x_n - x^*).$$

So the *error* $e_n = x_n - x^*$ obeys $e_{n+1} = (1 - \eta k)\, e_n$: a geometric sequence with ratio $1 - \eta k$. It decays to zero exactly when the ratio has magnitude below one,

$$|1 - \eta k| < 1 \qquad \Longleftrightarrow \qquad 0 < \eta < \frac{2}{k},$$

and it is fastest at $\eta = 1/k$, where the ratio is 0 and a single step lands exactly on x^*. For $\eta > 2/k$ the ratio exceeds one in magnitude, so the error grows and the iterates oscillate *outward* and diverge—the overshoot made precise. (For the parabola $f(x) = x^2$ above, $k = 2$, so the safe range is $0 < \eta < 1$ and $\eta = 0.3$ sits comfortably inside it.)

Remark (Choosing the step, and what it costs). The learning rate must therefore be chosen with care: too small and convergence crawls; too large—beyond $2/k$ for the model above—and the iterates overshoot the minimum and diverge. Gradient descent also finds only a *local* minimum—the one in whose valley it starts—so the starting point matters. With many parameters the single derivative becomes a *vector* of derivatives (the *gradient*), and the step is taken against it; the principle is unchanged.

Summary

Concept	Key idea
Tangent & normal	At $(a, f(a))$: tangent $y = f(a) + f'(a)(x - a)$; normal slope $-1/f'(a)$ (vertical if $f'(a) = 0$).
Intermediate value	Continuous f on $[a, b]$ hits every value between $f(a)$ and $f(b)$; a sign change forces a root.
Extreme value	Continuous f on a closed bounded $[a, b]$ attains an absolute max and min—so optimisation has an answer.
Mean value	Some c has $f'(c) = \dfrac{f(b) - f(a)}{b - a}$; hence $f' > 0 \Rightarrow$ increasing, $f' < 0 \Rightarrow$ decreasing, $f' \equiv 0 \Rightarrow$ constant.
Inverse derivative	$(f^{-1})'(b) = 1/f'(f^{-1}(b))$; gives $(\ln)' = 1/x$, $\arctan' = 1/(1 + x^2)$, $\arcsin' = 1/\sqrt{1 - x^2}$.
l'Hôpital	For $\frac{0}{0}$ or $\frac{\infty}{\infty}$, $\lim \frac{f}{g} = \lim \frac{f'}{g'}$; only for indeterminate forms.
Critical points	$f'(c) = 0$ or undefined; candidates for local extrema.
First-derivative test	f': $+ \to -$ max; $- \to +$ min; no sign change, neither.
Concavity	$f'' > 0$ concave up, $f'' < 0$ concave down; sign change of f'' is an inflection.
Second-derivative test	$f'(c) = 0$ with $f''(c) > 0$ min, $f''(c) < 0$ max, $f''(c) = 0$ inconclusive.
Curve sketching	Domain/symmetry, intercepts, asymptotes (vertical at denominator zeros, horizontal from limits at infinity), f' (extrema), f'' (concavity).
Optimisation	Reduce to one variable, solve $Q' = 0$, classify, check endpoints (extreme value theorem).
Gradient descent	$x_{n+1} = x_n - \eta f'(x_n)$: step against the derivative towards a minimum; η is the learning rate.

Next lecture: a change of direction—integration, the reverse of differentiation. We meet the definite integral as an area, the Fundamental Theorem of Calculus linking it to the derivative, and the method of substitution.

Companion material: a separate *Exercise Sheet 8* (with solutions) develops the practice problems for this unit.